Aker Solutions is an engineering and project execution company delivering solutions for the energy industry, including offshore structures, subsea systems, maintenance, modifications and energy transition projects.
In many of our fabrication projects, we design large steel structures made from standard profiles such as rectangular and square hollow sections. These components are connected by welds, and each weld needs to be identified, sized and calculated correctly.
The challenge is that this process is still largely manual. Engineers must interpret fabrication drawings, identify which components are connected, match them with the material list, and calculate weld thickness, length and volume for each connection. For large structures, this becomes time-consuming, repetitive and prone to inconsistency.
The goal of this case is to explore how AI can support this workflow. We want a solution that can read the drawing, detect welded connections, retrieve the relevant component data, and apply simplified engineering rules to calculate weld requirements automatically.

In short, the task is to move from manual weld interpretation and calculation toward a faster, more consistent and scalable engineering workflow.

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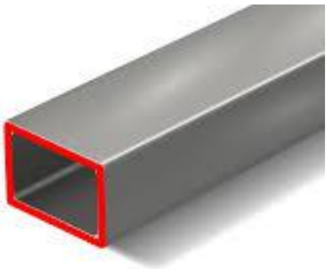
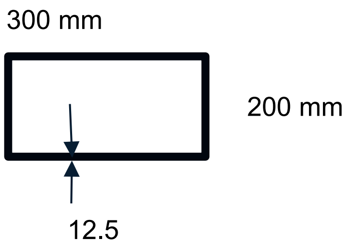
1 Background

In fabrication projects, welded steel structures are built from standard profiles such as:

- **RHS** (Rectangular Hollow Sections)
- **SHS** (Square Hollow Sections)



Example: RHS 300x200x12,5 has the following cross section with thickness 12,5 mm.



Each structure consists of multiple components connected by welds. These welds must be dimensioned and calculated according to engineering standards.

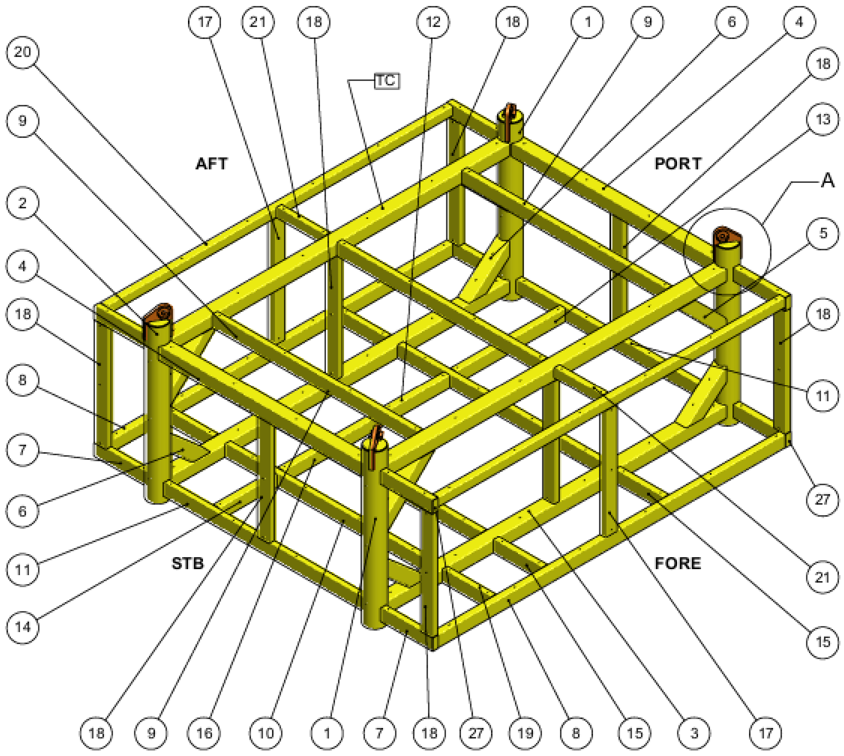
Today, much of this work is **manual and time-consuming**, requiring engineers to interpret drawings and calculate weld sizes and volumes for each connection.

2 The Challenge

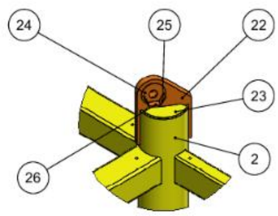
Develop an AI-based solution that can **automatically identify connections and calculate weld requirements** based on engineering drawings and material data

Participants will work with:

1. **fabrication drawing**
    - Each component is identified by a number
    - Connections between components must be interpreted visually



ISO VIEW



DETAIL A
SCALE 1 : 50

Material/part list (table)

- This table includes:
    - Profile type (e.g., RHS 400x200x12.5)
    - Dimensions

Length, material grade and weight can be disregarded for this hackathon case

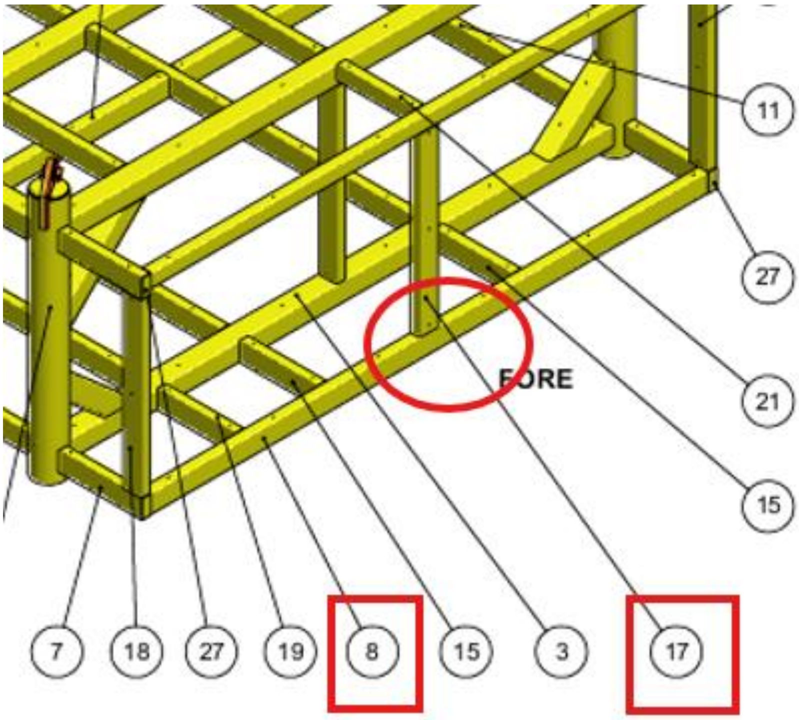
27	8	PLATE, 6		NORSOK M-120 Y05	3
26	8	PLATE, 10		NORSOK M-120 Y05	1
25	16	PLATE, 10		NORSOK M-120 Y05	1
24	8	PLATE, 10		NORSOK M-120 Y05	5
23	8	PLATE, 20		NORSOK M-120 Y05	15
22	4	PLATE, 100		NORSOK M-120 Y20	424
21	2	RHS300x200x12,5	1616	NORSOK M-120 Y07	145
20	2	RHS300x200x12,5	11533	NORSOK M-120 Y07	1038
19	1	RHS400x200x12,5	1616	NORSOK M-120 Y07	224
18	8	RHS400x200x12,5	3554	NORSOK M-120 Y07	389
17	2	RHS400x200x12,5	3654	NORSOK M-120 Y07	400
16	1	RHS400x200x16	1505	NORSOK M-120 Y07	209
15	5	RHS400x200x16	1616	NORSOK M-120 Y07	224
14	1	RHS400x200x16	1942	NORSOK M-120 Y07	270
13	1	RHS400x200x16	3647	NORSOK M-120 Y07	506
12	1	RHS400x200x16	3840	NORSOK M-120 Y07	533
11	2	RHS400x200x16	6644	NORSOK M-120 Y07	921
10	3	RHS400x200x16	6820	NORSOK M-120 Y07	947
9	3	RHS400x200x16	6820	NORSOK M-120 Y07	947
8	2	RHS400x200x16	11533	NORSOK M-120 Y07	1602
7	8	SHS400x400x16	1728	NORSOK M-120 Y07	239
6	4	SHS400x400x16	1761	NORSOK M-120 Y07	250
5	4	SHS400x400x16	2246	NORSOK M-120 Y07	342
4	2	SHS400x400x16	6759	NORSOK M-120 Y07	1257
3	4	SHS400x400x16	11272	NORSOK M-120 Y07	2104
2	2	CHS 610 x 25	5104	NORSOK M-120 Y07	1820
1	2	CHS 610 x 25	5104	NORSOK M-120 Y07	1820
ITEM NO.	QTY.	DESCRIPTION	LENGTH	MATERIAL GRADE	WEIGHT (kg)

3 Key Engineering Rule (Simplified)

For fillet welds:

- Weld thickness (**a**) =
  0.8 × thickness of the <u>thinner</u> connected part
- Weld length (**L**) =
  Perimeter of the profile cross-section
- Weld volume =
  a × a × L

4 EXAMPLE



ISO VIEW

The connection marked in red circle consists of member 8 and 17.

18	8	RHS400x200x12.5	3554	NORSOK M-120 Y07	389
17	2	RHS400x200x12,5	3654	NORSOK M-120 Y07	400
16	1	RHS400x200x16	1505	NORSOK M-120 Y07	209
15	5	RHS400x200x16	1616	NORSOK M-120 Y07	224
14	1	RHS400x200x16	1942	NORSOK M-120 Y07	270
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10	3	RHS400x200x16	6820	NORSOK M-120 Y07	947
9	3	RHS400x200x16	6820	NORSOK M-120 Y07	947
8	2	RHS400x200x16	11533	NORSOK M-120 Y07	1602

- Member 17: RHS 400×200×12.5
- Member 8: RHS 400×200×16

Steps:

1. Identify thinnest part → 12.5 mm
2. Calculate weld thickness:
   → a = 0.8 × 12.5 = 10 mm

3. Calculate weld length:
   → L = 400 + 400 + 200 + 200 = 1200 mm

4. Calculate weld volume:
   → $10 \times 10 \times 1200 = 120{,}000\ mm^3$

5 Task

Build a solution that can:

1. **Identify connections**
   - Detect which components are welded together from the drawing
2. **Retrieve component data**
   - Match component numbers to the table
   - Extract dimensions and thickness
3. **Apply engineering rules**
   - Determine the controlling (thinnest) thickness
   - Calculate weld size and volume
4. **Scale**
   - Perform calculations for all connections in the structure

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